

**Sample Paper**  
**FINALTERM EXAMINATION**  
Fall 2022  
CS302P – Digital Logic Design (Practical)

Time: 90 min  
Marks: 40

## MCQ Part

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**Question No: 1** (Marks: 01) - Please choose the correct option

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Generally, for checking the result in a single bit, we use \_\_\_\_\_ indicator.

- A. Logic Converter
- B. Logic Tester
- C. Probe**
- D. 7-segment Display

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**Question No: 2** (Marks: 01) - Please choose the correct option

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We can simplify a logic circuit using \_\_\_\_\_ device in Electronic Workbench.

- A. Truth Table
- B. Boolean Algebra
- C. Circuit Minimizer
- D. Logic Converter**

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**Question No: 3** (Marks: 01) - Please choose the correct option

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If  $Z = (A+B'+C)'$  then Z will be logic high only if \_\_\_\_\_.

- A.  $A = 1, B = 1, C = 1$
- B.  $A = 0, B = 0, C = 0$
- C.  $A = 0, B = 1, C = 0$**
- D.  $A = 0, B = 0, C = 1$

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**Question No: 4** (Marks: 01) - Please choose the correct option

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A logic circuit having N inputs, can have \_\_\_\_\_ different input combinations.

- A.  $N \times N$
- B.  $2 \times N$
- C.  $N+2$**

D.  $2^N$

**Question No: 5** (Marks: 01) - Please choose the correct option

In a 4-bit odd counter, if the current state is 0001 then the next state will be \_\_\_\_\_.

- A. 0010
- B. 0000
- C. **0011**
- D. 1100

**Question No: 6** (Marks: 01) - Please choose the correct option

The 4-bit grey code for the binary 1001 will be \_\_\_\_\_.

- A. 1100
- B. 0011
- C. 1111
- D. **1101**

**Question No: 7** (Marks: 01) - Please choose the correct option

What will be boolean expression for the function  $F(A, B, C) = \Pi(0, 1, 3, 7)$

- A.  $(A+B+C)(A+B+C')(A+B'+C')(A'+B'+C')$
- B.  $(A+B'+C)(A+B+C')(A'+B'+C)(A+B+C)$
- C.  $(A'+B'+C)(A'+B+C')(A'+B+C)(A'+B'+C')$
- D.  $(A'+B+C)(A+B'+C)(A+B+C')(A'+B'+C')$

**Question No: 8** (Marks: 01) - Please choose the correct option

Which of the following is the correct maxterm ID for the term  $x+y'+z$ ?

- A. M2
- B. M5
- C. m2
- D. **m5**

**Question No: 9** (Marks: 01) - Please choose the correct option

Which of the following gates is used to get the sum output of the full adder circuit?

- A. AND
- B. OR
- C. NOR
- D. **XOR**

**Question No: 10** (Marks: 01) - Please choose the correct option

To build an XOR gate using basic gates only, \_\_\_\_\_ gates can be used?

- A. 4 NAND
- B. 2 NAND
- C. 2 NOR
- D. 4 NOR

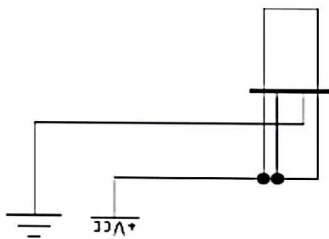
**Question No: 11** (Marks: 01) - Please choose the correct option

Which of the following is correct expression for carry output in a half-adder circuit?

- A.  $A \text{ XOR } B$
- B.  $A \text{ NOR } B$
- C.  $A \text{ AND } B$
- D.  $A \text{ NAND } B$

**Question No: 12** (Marks: 01) - Please choose the correct option

Consider the given circuit, what value will be displayed on 7-segment display when the simulation is activated?



- A. 3
- B. 12
- C. d
- D. 6

**Question No: 13** (Marks: 01) - Please choose the correct option

When using full adders to build a 4-bit parallel subtractor circuit, the carry input of which of the full adders is always set to 1?

- A. Full adder corresponding to least significant bit

- B. Full adder corresponding to most significant bit
- C. All 4 full adders are set to 1
- D. None of the full adders are set to 1

**Question No: 14** (Marks: 01) - Please choose the correct option

Which of the following is not a type of combinational circuit?

- A. Multiplexer
- B. Demultiplexer
- C. Counter
- D. Full Adder

**Question No: 15** (Marks: 01) - Please choose the correct option

Which of the following is not a type of sequential circuit?

- A. Counter
- B. Register
- C. Flip-flop
- D. Full Adder

**Question No: 16** (Marks: 01) - Please choose the correct option

In a 74151, 1-of-8 Data Sel/Mux, which of the following pins is used as strobe signal?

- A. Y
- B. W
- C. G'
- D. GND

**Question No: 17** (Marks: 01) - Please choose the correct option

In a 1-of-8 decoder, what will be the value of output Y when strobe pin is set to logic high?

- A. Y=0
- B. Y=W
- C. Y=1
- D. Y=D0

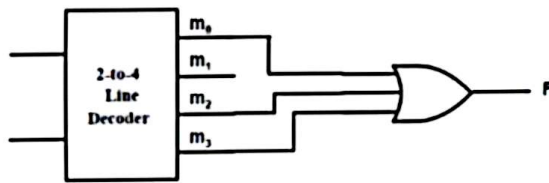
**Question No: 18** (Marks: 01) - Please choose the correct option

In 2-to-4 Line Decoder, which decoder output will be set to 1 for the binary inputs  $A = 1$  and  $B = 0$ ?

- A. D0
- B. D1
- C. D2
- D. D3

**Question No: 19** (Marks: 01) - Please choose the correct option

What will be boolean expression corresponding to below given decoder circuit?



- A.  $AB + AB' + A'B$
- B.  $AB + A'B'C + A'B$
- C.  $AB' + AB + A'B'C$
- D.  $AB + A'B' + ABC$

**Question No: 20** (Marks: 01) - Please choose the correct option

For which of the following input combination of ground lines of 74138 decoder, all the decoder outputs are set to 1?

- A.  $G2A' = X, G2B' = 1, G1 = X$
- B.  $G2A' = X, G2B' = X, G1 = X$
- C.  $G2A' = 1, G2B' = 1, G1 = X$
- D.  $G2A' = X, G2B' = X, G1 = X$

**Question No: 21** (Marks: 01) - Please choose the correct option

For which of the following combinations of ground input lines, all the output lines of 74138 decoder correspond to stored address 0?

- A.  $G2A' = X, G2B' = 1, G1 = X$
- B.  $G2A' = X, G2B' = X, G1 = X$
- C.  $G2A' = 1, G2B' = 0, G1 = 1$
- D.  $G2A' = X, G2B' = X, G1 = X$

**Question No: 22** (Marks: 01) - Please choose the correct option

To create a binary to gray code converter circuit by using the logic gates only, which of the following gates is used?

- A. AND
- B. OR
- C. NAND
- D. XOR

**Question No: 23** (Marks: 01) - Please choose the correct option

Consider the below given K-Map. The simplified expression corresponding to this K-map will be \_\_\_\_\_?

| yz \ wx | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      |    | 1  |    |    |
| 01      |    | 1  |    |    |
| 11      | 1  | 1  |    |    |
| 10      |    |    |    |    |

- A.  $W Y' Z + W' Y' Z$
- B.  $W X Y' + W' Y' Z$
- C.  $W Y' Z' + W Y' Z$
- D.  $W Y Z' + W Y Z$

**Question No: 24** (Marks: 01) - Please choose the correct option

Which of the examples below expresses the commutative law of multiplication?

- A.  $A + B = B + A$
- B.  $A \cdot B = B + A$
- C.  $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
- D.  $A \cdot B = B \cdot A$

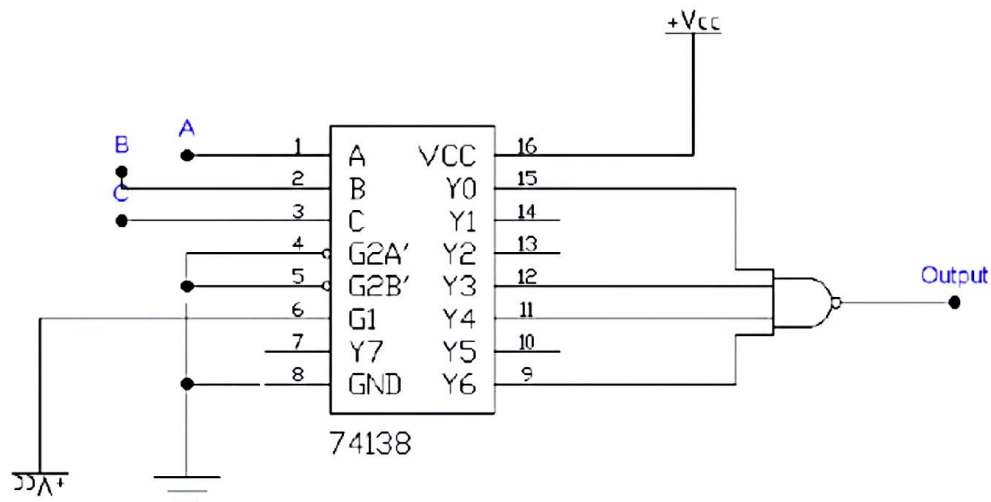
## Descriptive Part

**Question No: 25** (Marks: 05)

Consider the below given truth table corresponding to 3-input boolean function.  
Implement the given function in Electronics Workbench by using 74138 decoder.

| Inputs |   |   | Output |
|--------|---|---|--------|
| A      | B | C | F      |
| 0      | 0 | 0 | 1      |
| 0      | 0 | 1 | 0      |
| 0      | 1 | 0 | 0      |
| 0      | 1 | 1 | 1      |
| 1      | 0 | 0 | 1      |
| 1      | 0 | 1 | 0      |
| 1      | 1 | 0 | 1      |
| 1      | 1 | 1 | 0      |

**Solution:**

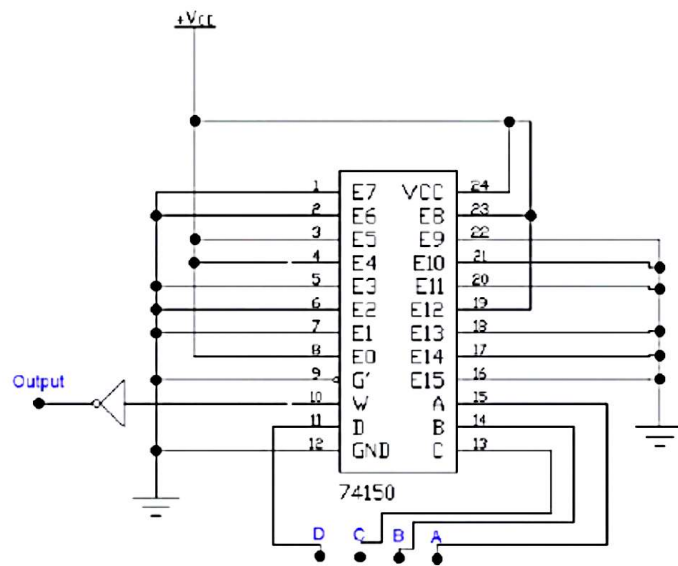


**Question No: 26 (Marks: 05)**

Realize the below given 4-variable function by using 16x1 multiplexer in Electronics Workbench.

$$F(A, B, C, D) = A'B'C'D' + A'BC'D + ABC'D' + A'BC'D' + AB'C'D'$$

**Solution:**



**Question No: 27 (Marks: 05)**

Simplify the given boolean function by using boolean algebra rules.

$$F = AB + (AC)' + AB'C(AB + C)$$

**Solution:**

$$\begin{aligned}
 F &= AB + (AC)' + AB'C(AB + C) \\
 &= AB + A' + C' + AB'C.AB + AB'C.C \\
 &= AB + A' + C' + 0 + AB'C \quad (B.B' = 0 \text{ and } C.C = C) \\
 &= ABC + ABC' + A' + C' + AB'C \quad (AB = AB(C + C')) \\
 &= ABC + ABC' = AC(B + B') + C'(AB + 1) + A' \\
 &= AC + C' + A' \quad (B + B' = 1 \text{ and } AB + 1 = 1) \\
 &= AC + (AC)' = 1
 \end{aligned}$$

**Question No: 28 (Marks: 05)**

Simplify the following Boolean expression using Karnaugh map:

$$F(w,x,y,z) = \Sigma(1,3,7,11,15) + x(0,2,5,8)$$

**Solution:**

The K-Map corresponding to given expression is as follows:

| WX/YZ | 00 | 01 | 11 | 10 |
|-------|----|----|----|----|
|-------|----|----|----|----|

|    |   |   |   |   |
|----|---|---|---|---|
| 00 | x | 1 | 1 | x |
| 01 |   | x | 1 |   |
| 11 |   |   | 1 |   |
| 10 | x |   | 1 |   |

There will be two minterm groups in the given K-Map as following:

| WX/YZ | 00 | 01 | 11 | 10 |
|-------|----|----|----|----|
| 00    | x  | 1  | 1  | x  |
| 01    |    | x  | 1  |    |
| 11    |    |    | 1  |    |
| 10    | x  |    | 1  |    |

The simplified expression will be  $f(w,x,y,z) = w'x' + yz$

**Question No: 29 (Marks: 05)**

Implement a 3-bit parallel subtractor circuit in the Electronics Workbench which subtracts the decimal 2 from decimal 5 and displays the result on a 7-segment display.

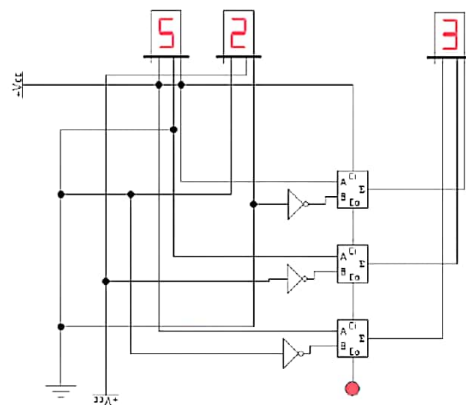
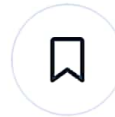
**Solution:**



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**Question No: 30 (Marks: 05)**

Build a 2-bit up counter in Electronics Workbench by using a D-Flop Flip.

**Solution:**